Week 4

Course Level Objectives:

- Recognize, examine, and interpret the basic principles of describing and presenting data. (CLO 2)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe the shape of a distribution (symmetric, left-skewed, right-skewed) from a dotplot or histogram. (CLO 2) **2.** Compute the mean and median of a dataset. (CLO 2) **3.** Explain how skewness and outliers affect the mean and the median (resistance). (CLO 2) **4.** Interpret the standard deviation as the typical distance from the mean. (CLO 2) **5.** Apply the 95% rule to estimate the mean and standard deviation from a bell-shaped dotplot. (CLO 2)	e-Text: 2.2 One Quantitative Variable Interactive Lecture: 2.2 One Quantitative Variable Slides: 2.2 One Quantitative Variable	Participation: Quantitative Variables – shape, mean vs. median; estimate the mean and SD of a dotplot without calculating (95% rule); Your Turn: Mean vs. Median with an Outlier (commute times) Participation: One Quantitative Variable (Kahoot!) – Wed: review of 2.2	Online Homework: 2.2 One Quantitative Variable (due Sun Sep 20, 11:59 PM) Take-Home Quiz 2 (due Mon Sep 21, 11:59 PM) – covers 2.1–2.3 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Kahoot! Respondus
6. Compute the sample standard deviation of a small dataset by hand. (CLO 2) **7.** Compute a z-score for a data value. (CLO 2) **8.** Use z-scores to compare values measured on different scales or in different groups. (CLO 2) **9.** Compute the five-number summary, range, and IQR of a dataset. (CLO 2) **10.** Apply the 1.5×IQR rule to identify outliers. (CLO 2) **11.** Construct a boxplot, marking any outliers. (CLO 2)	e-Text: 2.3 One Quantitative Variable: Using Percentiles Interactive Lecture: 2.3 One Quantitative Variable: Using Percentiles Slides: 2.3 One Quantitative Variable: Using Percentiles	Participation: Using Percentiles – Your Turn: Comparing Across Groups (heights of men and women with z-scores); five-number summary	Online Homework: 2.3 One Quantitative Variable: Using Percentiles (due Sun Sep 20, 11:59 PM) Take-Home Quiz 2 (due Mon Sep 21, 11:59 PM) – covers 2.1–2.3 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Respondus